Predictive Maintenance for Aircraft Engines: Machine Learning RUL Prediction

DSCI-441 M2 Final Project

Problem & Motivation

Unexpected engine degradation can cause costly delays, safety risk, and unnecessary maintenance.

Goal: estimate Remaining Useful Life (RUL) from current engine sensor data.

Output: predicted cycles remaining before failure.

Dataset

NASA C-MAPSS FD001 turbofan data.

100 training engines run to failure.

100 test engines with partial histories.

Each row: engine ID, cycle, 3 settings, 21 sensor readings.

Test labels come from RUL_FD001.txt.

Preprocessing

Training label: RUL = max_cycle - current_cycle.

Capped RUL at 130 cycles.

Removed constant / low-information features.

Retained informative settings and sensor channels.

Evaluated using final observed test cycle per engine.

Model Selection

Compared three regression models:

Ridge Regression baseline.

Random Forest for nonlinear sensor interactions.

Gradient Boosting as additional ensemble model.

Selection criterion: lowest RMSE on FD001 test labels.

Evaluation Results

Model	RMSE	MAE
Random Forest	18.20	13.18
Gradient Boosting	18.45	12.96
Ridge	21.23	16.93

Best model: Random Forest (RMSE = 18.20 cycles).

Deployment

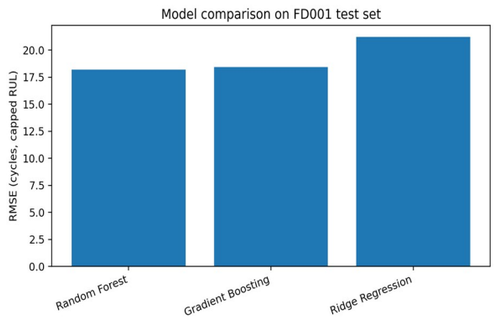
A Streamlit app was built for interactive predictions.

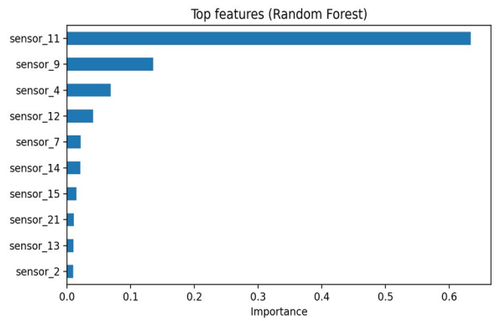
Upload CSV engine measurements.

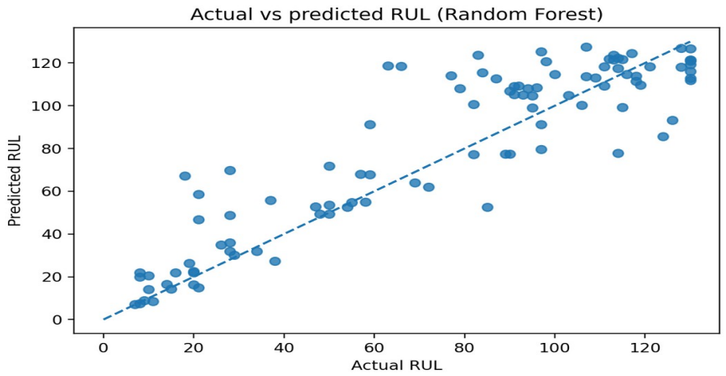
Or manually enter one current cycle.

App returns predicted RUL and maintenance priority.

File: app.py







Conclusion

Machine learning can convert multivariate engine sensor measurements into actionable RUL estimates. Random Forest performed best among tested models and is suitable for a clear, deployable predictive-maintenance prototype.

Data source: NASA C-MAPSS FD001 Turbofan Engine Degradation Simulation Data | Metrics: RMSE, MAE, R2 | Code includes model training, saved model, and Streamlit demo.